Catch timing, biology timing and selectivity

What two simulations say about opal's design

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1 The question

opal and Stock Synthesis (SS3), fitted to the same simulated Indian Ocean yellowfin data with the same growth, disagree about the stock:

opal configuration	B0 / SS3′s	Terminal SSB/B0
SS3 (reference)	1.00	0.399
A1: SS3′s age-based selectivity, fixed	1.03	0.404
C2: opal's length-based selectivity, estimated	0.88	0.307

With SS3′s selectivity opal reproduces SS3. With its own selectivity it finds a stock about 12% smaller and 0.1 more depleted. Three structural differences between the models could contribute, and each is a design choice for opal:

1. **Catch timing.** opal removes each quarter's catch instantaneously at the start of the quarter, then applies natural mortality. SS3 uses the Baranov equation, with fishing and natural mortality acting together.
2. **Biology timing.** opal evaluates the age–length key (which turns length-based selectivity into selectivity-at-age and predicts the length compositions) and the index at the start of the quarter. SS3 uses the middle of the season.
3. **Selectivity basis.** opal's selectivity is a function of length, reaching age through the age–length key. SS3′s curves for this model are functions of age.

Two simulations isolate these:

- Timing: catch removal (`start`, Pope's mid-step approximation, `baranov`) crossed with biology timing (`start`, `mid`).
- Selectivity: age-based against length-based selectivity, with opal's timing throughout.

2 Shared design

Both use one small model (`timing/R/timing-model.R`, R and RTMB) built from the IO yellowfin conversion:

- **Population:** quarterly, 256 quarters, ages 1–28 quarters (plus group); SS3′s M, maturity × fecundity and length-at-age; Beverton–Holt recruitment (steepness 0.8, σ_R 0.6); true B0 = C2′s, 4.10 million t.
- **Fisheries:** the 16 SS3 fleets pooled by gear into purse seine, longline and other, with their observed catches; selectivity from C2′s (or SS3′s) curves for one representative fleet each.
- **Data:** a longline index (CV 0.2, quarters 81–256), multinomial length compositions (ESS 20) for each fishery in every quarter with catch, catch known exactly.
- **Two kinds of test:**
  - *Deterministic:* the same parameters and catch projected under each convention. This is the structural effect on its own, and what matters for anything evaluated at fixed parameters (bridging runs, projections under a fixed catch).
  - *Simulation–estimation:* data from one convention, fitted under each. 30 replicates per study; 360 and 180 fits, all converged. Every model is compared with the correctly specified model on the same dataset, which cancels the bias they all share from recruitment-deviation shrinkage (B0 5–7% high, depletion about 0.03 low).

3 Results

3.1 Fixed parameters: conventions change depletion

Study	Configuration	Terminal SSB/B0	Difference
timing	start catch, mid biology	0.272	20%
timing	start catch, start biology	0.243	8%
timing	Baranov catch, mid biology	0.225	0%
timing	Pope catch, mid biology	0.222	−2%
timing	Baranov catch, start biology	0.195	−13%
timing	Pope catch, start biology	0.191	−16%
selectivity	age-based selectivity (SS3 curves)	0.250	
selectivity	its best length-based approximation	0.259	3.7%
selectivity	length-based selectivity (C2 curves)	0.243	
selectivity	its best age-based approximation	0.246	1.2%

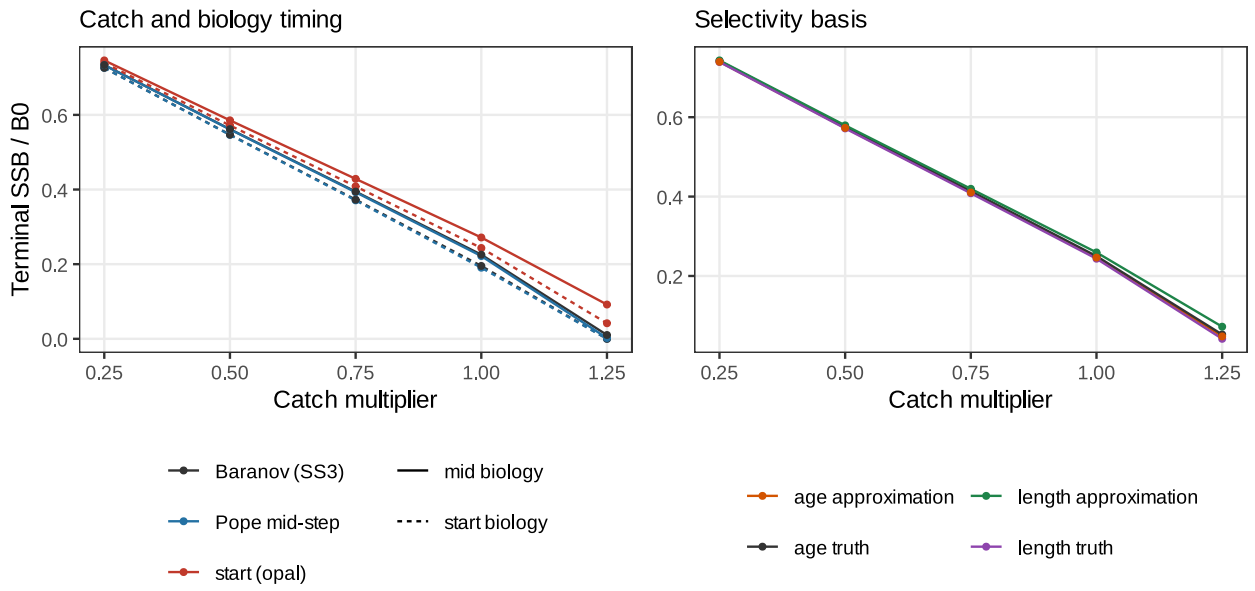


Figure 1: Terminal depletion against the catch multiplier, with identical parameters. Left: catch and biology timing. Right: each selectivity truth and its approximation in the other basis.

With the parameters held fixed:

- **Catch timing is the largest effect.** Start-of-quarter removal leaves the stock about 20–25% less depleted than Baranov for the same catch: a fish taken at the start is taken before natural mortality acts on it. The gap widens with the harvest rate, and near collapse Baranov fails where start-of-quarter removal does not.
- **Pope's mid-step approximation tracks Baranov** to within about 2%, without iteration.
- **Biology timing is about half as large and runs the other way**, so opal's start/start combination sits closer to SS3's Baranov/mid than either change alone.
- **Selectivity basis is small:** replacing a curve with its best approximation in the other basis moves depletion by 1–4%.

3.2 Fitted: conventions move B0, not depletion

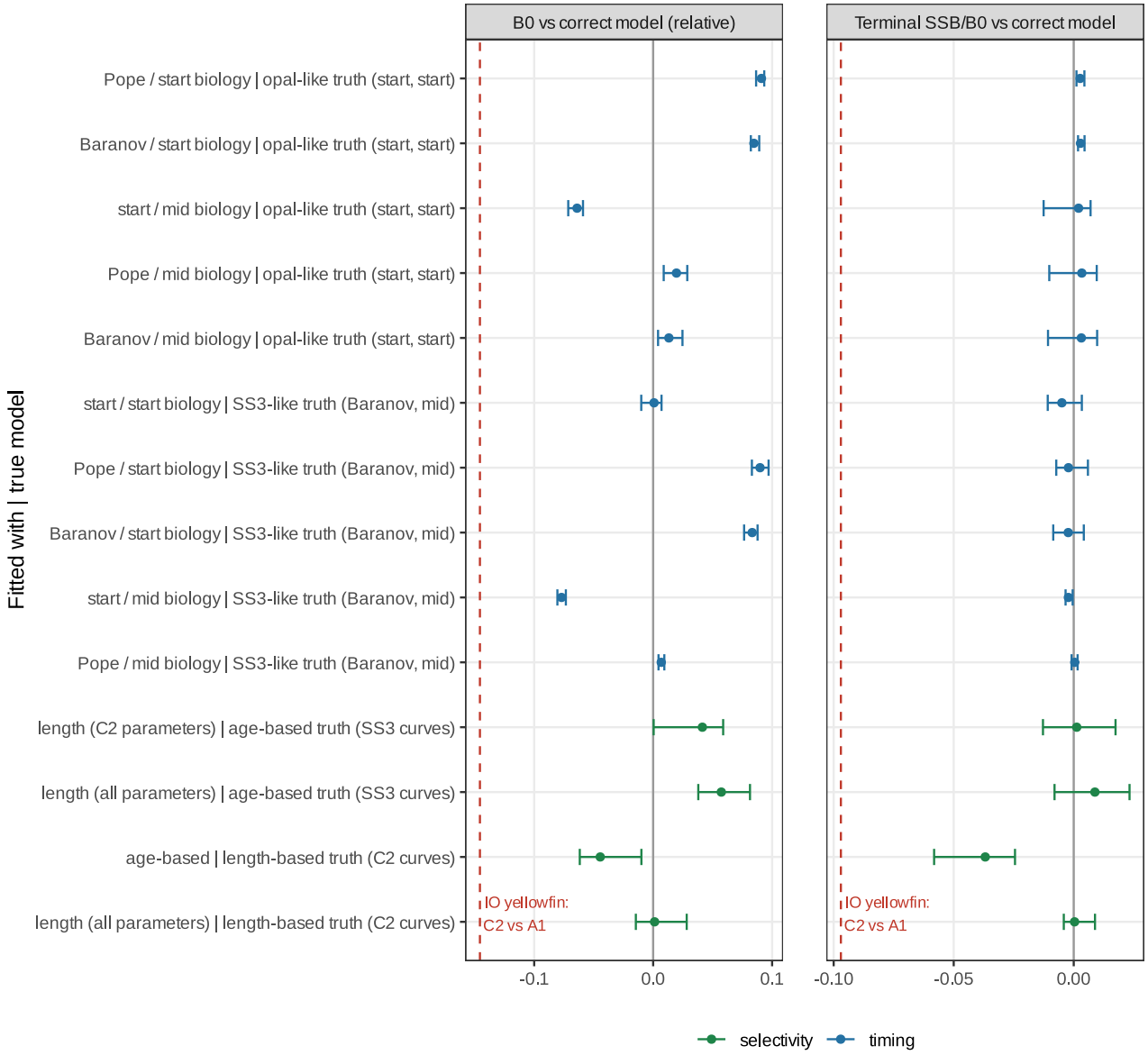


Figure 2: Effect of each misspecification on fitted B0 (left, relative to the correctly specified model) and terminal depletion (right, difference from it): medians and 10–90% ranges over 30 replicates. The dashed lines mark the IO yellowfin difference between C2 and A1 (opal's selectivity against SS3's).

Study	True model	Fitted with	B0 vs correct	Terminal dep. vs correct	10–90%
timing	opal-like	Pope / start	9.1%	0.003	0.001 to 0.004
timing	opal-like	Baranov / start	8.5%	0.003	0.002 to 0.005
timing	opal-like	start / mid	−6.4%	0.002	−0.013 to 0.007
timing	opal-like	Pope / mid	2.0%	0.003	−0.010 to 0.010
timing	opal-like	Baranov / mid	1.3%	0.003	−0.011 to 0.010
timing	SS3-like	start / start	0.1%	−0.005	−0.011 to 0.003
timing	SS3-like	Pope / start	9.0%	−0.002	−0.007 to 0.006
timing	SS3-like	Baranov / start	8.3%	−0.002	−0.009 to 0.004

Study	True model	Fitted with	B0 vs correct	Terminal dep. vs correct	10–90%
timing	SS3-like	start / mid	−7.7%	−0.002	−0.003 to −0.000
timing	SS3-like	Pope / mid	0.7%	0.000	−0.001 to 0.002
selectivity	age-based	length (C2 parameters)	4.1%	0.001	−0.013 to 0.017
selectivity	age-based	length (all parameters)	5.7%	0.009	−0.008 to 0.023
selectivity	length-based	age-based	−4.4%	−0.037	−0.058 to −0.024
selectivity	length-based	length (all parameters)	0.1%	0.000	−0.004 to 0.009

Once a model is fitted to an index and length compositions:

- **Timing misspecification goes almost entirely into B0.** Each timing convention moves B0 by about 7–9%, but terminal depletion by 0.005 or less. Because catch and biology timing push in opposite directions, opal's start/start fitted to SS3-like data lands within about 1% of the correct B0.
- **Selectivity misspecification is asymmetric.** A length-based model fitted to age-based data estimates B0 4–6% high and depletion correctly. An age-based model fitted to length-based data is biased low in depletion (about −0.04) and B0 (about −4%): it cannot represent larger fish within an age being caught more often.
- **None of them reproduces the IO yellowfin gap.** C2 against A1 is −15% in B0 and −0.097 in depletion. The largest simulated depletion effect is about −0.04, from the opposite misspecification (fitting age-based curves to length-based truth). Fitting length-based curves to SS3′s age-based truth, the situation closest to C2, leaves depletion unchanged and moves B0 *up*, not down.

4 What explains the IO yellowfin gap?

Not catch timing, biology timing or selectivity basis on their own, at least in the simplified setting. What the simulations leave out, and so the remaining candidates:

1. **All 16 fleets and their individual shapes.** Pooling into three fisheries with representative curves averages away fleets that a length-based double normal fits poorly in C2: fishery 5′s shallow ascending limb and fishery 9′s drop at the plus group.
2. **The survey's size-frequency data.** SS3 fitted the survey's generalised size compositions ($\lambda = 0.51$); the converter omits them, so in opal the index selectivity is informed only by the index.
3. **opal's N(0, 2) prior on selectivity parameters.** Nine of C's parameters sat at the prior mean. C2 showed fixing them makes no difference, but the prior also acts on the parameters the data inform.
4. **A spatial truth.** The data come from a 159-cell spatial operating model; pooled area fleets need not follow any single selectivity basis, and neither assessment is correctly specified.
5. **The interaction of misspecifications.** Each simulation changes one thing at a time. In C2 everything differs at once, with the real data's weighting.

5 Implications for opal

1. **Keep length-based selectivity as the default.** It is the more robust basis: unbiased in depletion when the truth is age-based, whereas an age-based model is biased when the truth is length-based. An age-based option would help reproduce SS3 models exactly, but status estimates do not need it.

2. **Treat catch and biology timing as a pair.** Their effects largely cancel. Adopting only one of SS3′s conventions would move opal's B0 about 8% further from SS3′s than it is today.
3. **Add Pope's mid-step catch as an option.** It matches Baranov within about 2%, needs no iteration and costs the same as opal's current method. With mid-step length-at-age and mid-step index abundance as companion options, opal would have an SS3-compatible timing mode for bridging and comparison work.
4. **Expect timing to matter where nothing is re-estimated.** Bridging runs at fixed parameters, projections under fixed catches, catch-based reference points and absolute biomass all carry the 20–25% deterministic difference, and more for annual time steps or higher harvest rates. Start-of-step removal is the most forgiving near collapse.
5. **Report B0 alongside depletion when comparing platforms.** Both studies show structural differences moving B0 by up to about 9% while depletion stays put; a comparison on depletion alone would miss them.

6 Next steps

- Rerun the selectivity design with all 16 fleets and their C2 curves, to test candidate 1.
- Refit C2 with the survey's size-frequency data converted (aggregating predicted compositions to the coarse bins), to test candidate 2.
- Refit C2 with the selectivity prior widened or removed, to test candidate 3.
- If opal adopts Pope mid-step catch or mid-step biology, rerun the IO yellowfin bridging check at SS3′s parameters, which should then track SS3 closer than the current 10%.

7 Caveats

Both simulations use three pooled fisheries, known growth, M and steepness, time-invariant selectivity, and quarterly steps with modest harvest rates (at most about 0.2 per quarter). Fits start at the true values (`nlminb` stalls from zero recruitment deviations). Results are paired against the correctly specified model, which shares a bias from recruitment-deviation shrinkage with every other model.

8 Reproducing

Study	Scripts	Time on 4 cores
Timing	`timing/01-om-inputs.r`, `02-deterministic.r`, `03-sim-est.r 30 4`	41 min
Selectivity	`selectivity/01-om-inputs.r`, `02-deterministic.r`, `03-sim-est.r 30 4`	16 min
This document	`sims-synthesis.qmd`, reads both `data/` folders	seconds

Run the timing study's step 1 first: the selectivity study reuses its inputs. In a fresh cloud container, `code/rtmb/setup/install-r-opal.sh` installs R, RTMB and opal.